

Zhibo Wang

COMPUTER ENGINEERING | EMBEDDED SYSTEMS | DIGITAL HARDWARE
| EDGE AI

La Jolla, California
(+1) 415-660-8327 | zhw112@ucsd.edu
github.com/Burger0328

PROFESSIONAL SUMMARY

Computer Engineering graduate (August 2026) and current AI Research Intern with hands-on experience across embedded systems, FPGA and RTL design, TinyML, VLSI verification, power electronics, performance engineering, and full-stack software. Builds complete systems from sensing and circuit integration through firmware, validation, data pipelines, dashboards, deployment, and physical prototyping. Brings professional troubleshooting experience and clear bilingual communication.

EDUCATION

University of California, San Diego

B.S. Computer Engineering | La Jolla, California

September 2024 - August 2026

TECHNICAL SKILLS

Programming	Python, C, C++, SystemVerilog/Verilog, TypeScript, HTML, CSS
Embedded / IoT	ESP32-S3, sensor integration, MQTT, TensorFlow Lite Micro, microSD, camera systems
Digital / Circuits	FPGA architecture, RTL verification, CMOS logic, DRC/LVS, HSPICE, power converters, MPPT
Software / Data	React, FastAPI, MySQL, SQLite, Docker, Git, MCP, AI Skills, NumPy, SciPy, OpenCV, WebSockets, automated testing
Engineering Tools	Quartus, Cadence Virtuoso, Altair PSIM, SolidWorks, CAD, GitHub Actions
Hands-on	Circuit assembly, hardware debugging, 3D-printed enclosures, PC repair, router and network setup

PROFESSIONAL EXPERIENCE

AI Research Intern | Shenzhen Yansi Intelligent Technology Co., Ltd.

September 2026 - Present

- Contribute to Science Intelligent Research Radar, a React 19 / TypeScript and FastAPI platform with AI Skills workflows and an MCP server for public-literature analysis, citation verification, and structured research actions.
- Implemented an isolated Docker Compose deployment path and main-branch GitHub Actions server workflow; documented environment setup and release procedures for repeatable team delivery.

Technical Support | Chinese Bay Services

San Francisco, California | January 2024 - Present

- Support the hardware, software, and network systems used in daily tax-service operations.
- Diagnose and repair desktop and laptop failures, including hardware, circuit, and peripheral issues.
- Configure routers, restore local connectivity, document fixes, and guide users through reliable recovery steps.

Technical Support | Pacific East Mall

Richmond, California | September 2022 - January 2023

- Provided on-site customer support for software, operating-system, and general computer issues.
- Translated user-reported symptoms into repeatable tests and explained solutions in clear, practical language.

ENGINEERING PROJECTS

SniffKitty Odor Sensor

ESP32-S3 | C++ | MQ135 / MQ137 | AHT10 | Camera | microSD | CAD

- Built six working hardware variants combining gas and environmental sensing, camera-based presence detection, local logging, and a browser dashboard.
- Implemented startup baseline calibration, Wi-Fi access-point fallback, event-photo capture, storage recovery, and a ventilated 3D-printed enclosure.

Thermal Presence TinyML System

ESP32-S3 | AMG8833 | TensorFlow Lite Micro | FastAPI | MQTT | MySQL | Docker

- Engineered a 76-feature thermal pipeline and deployed a 6.7 KB INT8 neural network with 89.04% accuracy on 3,513 group-held-out readings.
- Delivered local inference plus MQTT messaging, authenticated FastAPI services, MySQL persistence, WebSocket updates, and a live 8 x 8 heatmap.

ENGINEERING PROJECTS (CONTINUED)

Renewable Energy MPPT Control

Altair PSIM | P&O; MPPT | Interleaved Buck-Boost | PI Control | PWM

- Designed layered control for solar and wind conversion with three 120-degree interleaved converter legs and perturb-and-observe tracking.
- Reached 2,248.7-2,249.9 W against a 2,250 W solar maximum and demonstrated rapid adaptation across multiple wind speeds in simulation.

FPGA Viterbi Decoder

SystemVerilog | Python | Quartus | Cyclone V

- Implemented a synthesizable rate-1/2 convolutional encoder and eight-state hard-decision decoder with 64-symbol traceback.
- Created a Python reference model, reproducible error experiments, and a self-checking RTL testbench; completed fitting at 559 ALMs and 579 registers.

CPU Performance Engineering

C++ | OpenMP | x86 Assembly | Hardware Counters | GitHub Actions

- Connected execution time to CPI, instruction count, cache and TLB locality, branch behavior, false sharing, synchronization, and Amdahl's Law.
- Measured a 13.77x row-major locality speedup, 3.42x compiler speedup, and a cache-aware hit-rate increase from 14.24% to 93.73%.

Digital Image Processing Toolkit

Python | NumPy | SciPy | OpenCV | PyWavelets | pytest

- Refactored six experiments into an installable package covering transforms, Fourier filtering, restoration, compression, morphology, and segmentation.
- Added validation, deterministic controls, numerical reconstruction tests, a standalone visual demo, and continuous integration.

CMOS VLSI Design and Verification

Cadence Virtuoso | HSPICE | CMOS | DRC | LVS

- Designed and characterized transistor-level gates, transmission-gate storage, flip-flops, and a registered consecutive-ones detector.
- Completed DRC/LVS and reported a 49.31 ps minimum passing period, 68.8 ps detector period, and 398.689 uW simulated power.

House of Odds

TypeScript | Vite | Konva | Prisma | SQLite | Vitest

- Implemented the final-results model, controller, and view for a three-act probability-learning game, including per-act accuracy and restart behavior.
- Added regression coverage and integration fixes; the repository contains 38 passing tests across eight files.

ADDITIONAL

Languages	Chinese (native) English (fluent)
Strengths	End-to-end prototyping Verification-first development Hardware-software integration Technical communication